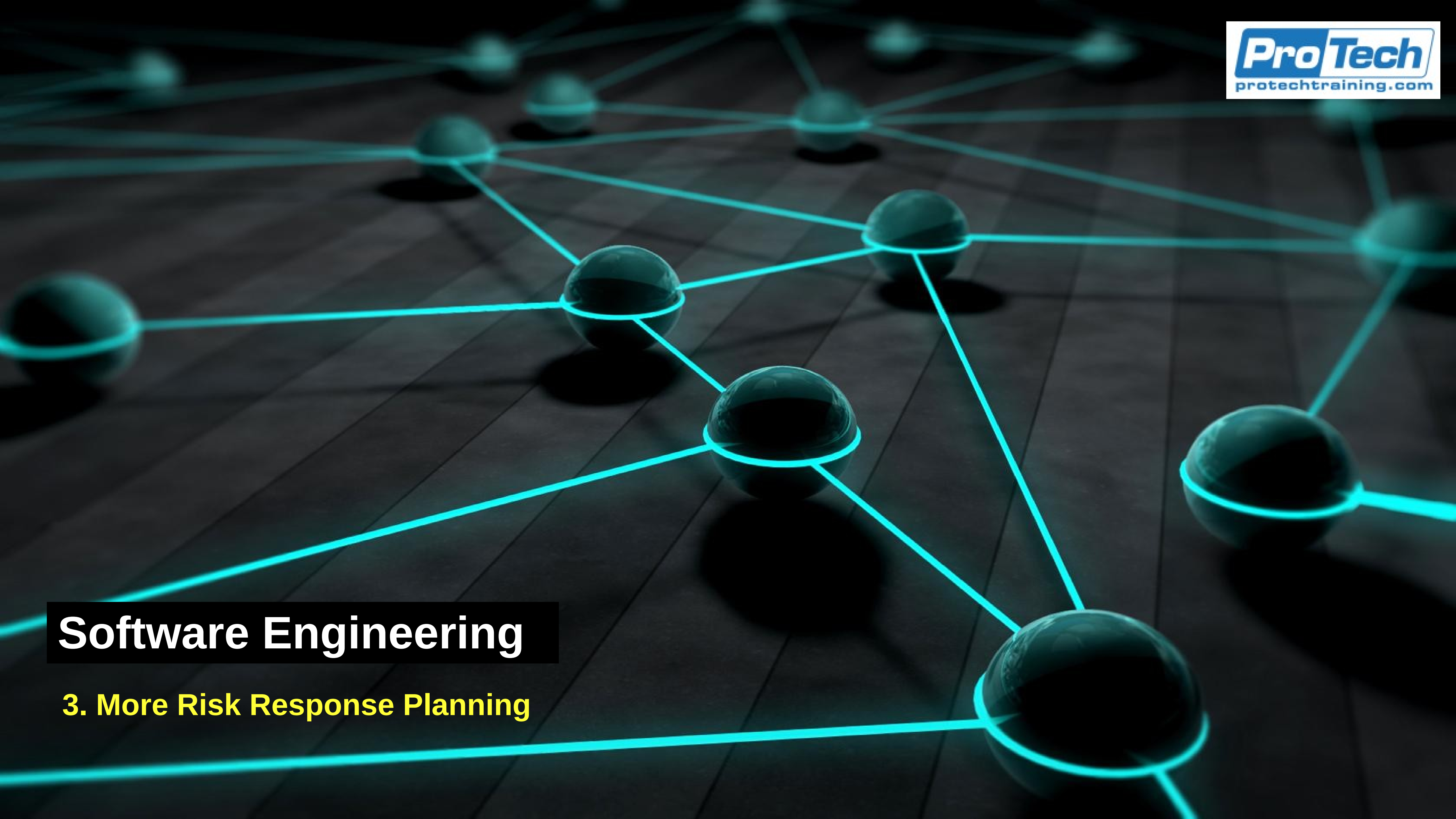

Software Engineering

3. More Risk Response Planning

Mitigation Planning

- Mitigation planning
  - How an organization turns abstract risk analysis into concrete, executed change
  - Risk assessment describes what could go wrong
  - Risk response chooses how to treat it (avoid/mitigate/transfer/accept)
- The mitigation plan answers
  - “Exactly what will we do, by when, with which resources, and how will we know we’ve succeeded?”
- In mature environments
  - Mitigation plans are tightly integrated with
    - *Risk registers, GRC workflows, project management, and operational metrics (KPIs/KRIs)*
    - *Poorly defined mitigation is one of the most common failure modes in risk program*
    - *Risks get “accepted by default” because mitigation is vague, under-resourced, or untracked*

Mitigation Planning

- A robust mitigation plan has four pillars
  - Resources
  - Timelines
  - Success criteria
  - Clear ownership

Resource Planning

- Mitigation without adequate resources is performative
  - It exists only on paper
  - Resource planning ensures the organization has the resources to implement the plan
- Technical resources
  - These are the people who actually change things in systems
    - *Engineers / Developers: apply code changes, refactor applications, improve error handling*
    - *Platform / Cloud Engineers: adjust infrastructure, implement auto-scaling, reconfigure IAM*
    - *Security Analysts / Engineers: implement new controls, tuning SIEM/SOAR rules, deploy EDR*
    - *Architects: redesign critical parts of the system (e.g., remove single points of failure, break monoliths, segment networks)*

Resource Planning

- Technical resources
  - Example
  - Mitigation: “Reduce lateral movement risk in internal network”
  - Technical Resources
    - *Network engineer to implement segmentation*
    - *Security engineer to tune microsegmentation policy*
    - *Architect to define zero-trust design patterns*

Resource Planning

- Financial resources
  - Mitigation often requires spend, including
    - *New tools (e.g., CSPM platform, PAM solution, backup system)*
    - *Additional cloud capacity or redundancy*
    - *External consultants or penetration testers*
    - *Training programs for staff*
  - Risk teams must coordinate with Finance and leadership to secure budget
  - Expenditures often justifying it with evidence for
    - *Risk-reduction value*
    - *Cost of inaction vs. cost of mitigation*
    - *Regulatory/penalty avoidance*

Resource Planning

- Human resources: owners and SMEs
  - Mitigation needs named people, not just roles
    - *Risk Owner: accountable for risk closure and residual risk*
    - *Mitigation Owner / Project Lead: coordinates the actual remediation activities*
    - *Subject Matter Experts (SMEs): cryptography, identity, DB, cloud, etc.*
  - High-maturity organizations track workload to ensure people tasked with mitigation actually have capacity, not just theoretical assignment

Resource Planning

- Infrastructure resources
  - Mitigation often requires safe testing and rollout environments, including
    - *Non-production environments that resemble production (staging, pre-prod)*
    - *Sandbox environments for testing patches, configuration changes, and failover*
  - Extra cloud capacity for
    - *Blue/green deployments*
    - *Rolling updates*
    - *DR testing*
    - *Parallel runs*

Resource Planning

- Infrastructure resources
  - Example:
  - Mitigation: “Implement real-time database replication across regions”
  - Required infrastructure
    - *Additional DB nodes in second region*
    - *Replication network setup*
    - *Load balancer configuration*
  - Without infrastructure planning, mitigation may raise new risks
    - *For example: untested failover, performance degradation*

Timelines

- Timelines transform mitigation
  - From “we should do this”
  - Into “we will do this by X, and we will measure progress along the way”
- Risk-based timelines
  - No all risks are equal.
  - Timeline design must consider
    - *Inherent risk level (likelihood × impact)*
    - *Exposure duration – how long the organization has been exposed*
    - *Risk velocity – how fast incidents could materialize*
    - *Compensating controls – can we “buy time” safely?*
  - Critical, high-velocity risks require days, not months

Timelines

- Regulatory alignment
  - Certain timelines are effectively non-negotiable due to law or regulation
    - *Patching deadlines for critical vulnerabilities in regulated industries*
    - *Time-bound remediation of audit findings*
    - *Contractual remediation SLAs with clients*
  - Mitigation plans must explicitly reference these obligations and prioritize accordingly
  - Example
    - *“Remediate high CVSS vulnerabilities within 30 days, critical within 7 days”*
    - *“Address audit-identified high-severity control deficiencies within 90 days”*

Timelines

- Explicit milestones
  - Rather than a single end-date, mature plans define intermediate milestones, such as
    - *Design complete by Week 2*
    - *Test environment implemented by Week 4*
    - *Pilot deployment completed by Week 6*
    - *Full production rollout by Week 8*
    - *Verification and closure by Week 10*
  - Milestones allow
    - *Progress tracking*
    - *Early detection of delays*
    - *Management visibility*
  - This is where risk mitigation intersects with project management discipline

Success Criteria

- Define what "done" means
  - Whether risk has actually been reduced, not just "activities performed"
  - Good success criteria are
    - *Specific: tied to a defined metric or state*
    - *Measurable: quantifiable with data*
    - *Auditable: evidence can be produced*
    - *Risk-linked: clearly reduce likelihood/impact*

Success Criteria

- Example 1
  - “Reduce critical vulnerability count by 80% within 60 days”
    - *Metric: # of critical vulnerabilities from scanner X*
    - *Baseline: 500 critical vulns today*
    - *Target: 100 or fewer in 60 days*
    - *Risk Link: Direct reduction of attack surface*
    - *Auditability: Before/after scan reports*

Success Criteria

- Example 2
  - “Achieve 100% MFA enforcement on privileged accounts.”
    - *Metric: % of privileged identities requiring MFA.*
    - *Scope: Admin accounts, root users, domain admins, cloud privileged roles.*
    - *Target: 100%. No exceptions unless formally approved and documented.*
    - *Risk Link: Reduces credential-based compromise risk.*
    - *Auditability: Identity provider logs, configuration evidence.*

Success Criteria

- Example 3
  - “Reduce MTTR from 4 hours → 1 hour”
    - *Metric: Mean Time to Recover (MTTR) for SEV-1 incidents*
    - *Baseline: 4 hours average over last quarter*
    - *Target: 1 hour in next quarter*
  - Mitigation components
    - *Automation (SOAR, scripts)*
    - *Better on-call coverage*
    - *Improved diagnostics and runbooks*
    - *Risk Link: Limits impact duration and financial loss.*
    - *Auditability: Incident logs & metric dashboards.*

Success Criteria

- Example 4
  - “Zero backup integrity failures for 90 days”
    - *Metric: # of failed restore tests or corrupt backups*
    - *Baseline: e.g., 5 integrity failures in past 90 days*
    - *Target: 0 failures in the next 90 days*
  - Mitigation components
    - *Better backup procedures*
    - *Verification scripts*
    - *Storage configuration fixes*
    - *Risk Link: Reduces data loss and recovery risk*
    - *Auditability: Backup reports and restore test logs*

Success Criteria

- Why success criteria matter
  - Without solid success criteria
    - *Mitigation degenerates into “activity theater”*
    - *Risk doesn’t actually change even though teams are busy*
    - *Leadership can’t tell if investments are reducing exposure*
  - With clear criteria, risk and leadership can evaluate
    - *Was the risk meaningfully reduced?*
    - *Did we hit deadlines?*
    - *Do we accept residual risk or plan further mitigation?*

Responsible Parties (RACI)

- If everyone is responsible, no one is responsible
  - Clear ownership is critical under pressure and for sustained follow-through
  - The RACI model in mitigation
    - *Responsible (R) – Does the work*
    - *Accountable (A) – Ultimately answerable; approves completion*
    - *Consulted (C) – Provides expertise, must be engaged*
    - *Informed (I) – Kept updated; doesn't influence day-to-day actions*

Responsible Parties (RACI)

- Responsible
  - *Ops engineer implementing failover*
  - *Security engineer deploying new EDR rules*
  - *Cloud engineer reconfiguring IAM or network*
- Accountable
  - *Product or system owner (e.g., Head of Payments Platform)*
  - *Ultimately “owns” the risk and signs off that mitigation is complete*
- Consulted
  - *Security team (control design and validation)*
  - *Risk and GRC (alignment with risk appetite and frameworks)*
  - *Legal and compliance (for regulatory considerations)*
- Informed
  - *Senior leadership (for major risk items)*
  - *Internal audit (for future assurance work)*
  - *Board-level committees (for strategic risks)*

Responsible Parties (RACI)

- Example: RACI for a Critical Vulnerability Mitigation
  - Risk: Critical remote code execution vulnerability in externally exposed web app
  - R – Responsible
    - *App dev team (apply fix)*
    - *DevOps team (deploy patch)*
  - A – Accountable
    - *Application owner (business owner of that service)*
  - C – Consulted
    - *Security engineering (validate patch effectiveness)*
    - *Architecture team (ensure patch aligns with standards)*
  - I – Informed
    - *CISO, CIO*
    - *Risk committee*

High-Quality Mitigation Plan

- Example
- Risk: Elevated credential theft risk due to lack of MFA on privileged accounts.
  - Strategy: Mitigate via enforcing MFA and reducing unnecessary privileged accounts.
  - Resources:
    - *Technical: IAM engineer, security engineer, directory services specialist*
    - *Financial: Budget for MFA token licensing*
    - *Infrastructure: Directory sync testing environment*
  - Timeline:
    - *Design & pilot: 2 weeks*
    - *Rollout to admins: 4 weeks*
    - *Rollout to service accounts where possible: 6 weeks*
  - Success Criteria:
    - *100% of human privileged accounts protected by MFA within 6 weeks*
    - *0 unprotected privileged logins detected in SIEM over 30-day verification period*
  - RACI:
    - *R: IAM engineer & security engineer*
    - *A: Head of Infrastructure*
    - *C: CISO, Security Architect*
    - *I: CIO, Internal Audit*

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Fallback / Contingency Plans

- Mitigation plans are designed to reduce risk by addressing root causes
  - But in real-world systems, especially complex, distributed, and tightly coupled ones, mitigation may
    - *Take longer than expected*
    - *Only partially succeed*
    - *Be blocked by external constraints (vendors, regulations, dependencies)*
    - *Introduce new or unforeseen side effects*
  - Fallback and contingency plans provide the safety net when mitigation is insufficient or too slow.
  - Core components of resilience engineering
    - *They ensuring that critical services continue to operate, perhaps in degraded form, even when normal controls or mitigation paths are compromised*

Fallback / Contingency Plans

- Operational workarounds and alternate pathways
  - Fallback plans are tactical
  - Short to medium term responses
  - Allow the organization to maintain some level of service despite failures in primary systems or processes
  - Assumes that
    - *The primary path is temporarily unavailable or unreliable*
    - *Full root-cause resolution will take time*
    - *The organization must continue operating in the meantime*
  - Fallbacks are deliberately designed into architectures and processes

Fallback / Contingency Plans

- Failover to backup systems
  - Failover mechanisms reroute workloads from a failing primary system to a secondary one
  - Examples
    - *Database failover from primary region to a hot standby in another region*
    - *Switching from primary message queue cluster to a backup cluster*
    - *Moving traffic from one data center to a mirrored facility*
  - Design considerations
    - *Data consistency: Is the backup RPO acceptable (e.g., 5 minutes of data loss)?*
    - *Performance: Can the backup handle full production load?*
    - *Automation: Is failover automatic (health checks, DNS failover) or manual (runbook-driven)?*
  - Failover is often a planned fallback, tested through BC/DR exercises

Fallback / Contingency Plans

- Switch from automated to manual processing
  - When automated systems fail or become unreliable
  - Manual processes can serve as temporary bridges
  - Examples
    - *Manual approval of high-value transactions when rule engines or fraud systems are offline*
    - *Customer service agents manually entering orders when front-end portals fail*
    - *Operations teams manually running batch jobs that normally run via scheduler*
  - Risks and constraints
    - *Human capacity limits (cannot scale indefinitely)*
    - *Higher error rates*
    - *Extended processing times*
  - Fallback planning must define how long manual mode is sustainable and what throughput is realistic

Fallback / Contingency Plans

- Reroute traffic to alternate regions
  - In cloud and distributed systems, fallback often involves redirecting traffic
  - Examples
    - *Routing users in Region A to Region B during an outage*
    - *Using global load balancers to steer traffic away from failing zones*
    - *Redirecting authentication to a secondary identity cluster*
  - Challenges
    - *Data residency/privacy constraints (e.g., cross-border data flow)*
    - *Latency increases affecting user experience*
    - *Capacity planning in alternate regions*
  - Fallback here is often tightly coupled with multi-region architecture and BC/DR strategies

Contingency Plans

- Fallback plans are more operational and tactical
- Contingency plans are strategic, high-level plans
  - For dealing with severe, persistent, or systemic failures when
    - *Primary and backup options are compromised*
    - *External dependencies fail dramatically (e.g., vendor outage, cloud provider failure)*
    - *Safety, legal, or regulatory concerns override normal operations*
- Contingency plans often involve major shifts in how or where operations are conducted

Contingency Plans

- Activate alternative suppliers or vendors
  - If a critical third-party becomes unavailable or unreliable
  - Contingency may involve shifting to alternatives
  - Examples
    - *Switching payment processor due to an extended outage*
    - *Moving from a compromised cloud security provider to another vendor*
    - *Activating a backup logistics company in a supply chain disruptio.*
  - Prerequisites
    - *Pre-negotiated contracts or frameworks*
    - *Tested integration path*
    - *Risk assessments of alternative providers*
  - This ties directly into third-party risk management and vendor resilience

Contingency Plans

- Transition operations to a disaster recovery (DR) site
  - When primary environments are severely compromised, contingency may involve full DR activation
  - Examples
    - *Moving core banking systems to a DR data center after a catastrophic event*
    - *Running all workloads from an alternate region for an extended period*
    - *Operating in DR mode after a ransomware event while restoring primary systems*
  - Key issues
    - *DR environments must be kept in sync (data, config, security posture)*
    - *Staff must be trained to operate from DR sites*
    - *Operating for extended periods in DR may incur substantial costs*
  - This is where BC/DR and contingency strategy fully overlap

Contingency Plans

- Switch authentication to offline mode
  - In identity crises (e.g., IdP outage, SSO failures, compromised SSO integration) may require some sort of contingency plan for authentication
  - Examples
    - *Temporarily allow cached or local authentication with strict constraints*
    - *Use one-time codes or alternate identity verification channels (e.g., helpdesk-initiated)*
    - *Limit access to only those who can authenticate via a secondary method*
  - Risk trade-off
    - *Security risk increases (less robust authentication)*
    - *Operational risk decreases (users can log in)*
    - *Contingency decisions here must be risk-informed and time-bound*

Contingency Plans

- Execute emergency “Stop” procedures for unsafe conditions
  - Sometimes, the safest action is to halt operations
  - Examples
    - *Stopping trading due to severe pricing or risk system malfunction*
    - *Pausing critical industrial processes in energy or manufacturing when safety systems are degraded*
    - *Halting high-value or high-risk transaction types (e.g., international wires) until controls are restored*
  - This is akin to a “kill switch” and must be
    - *Pre-authorized at the right level of leadership*
    - *Clearly documented and rehearsed*
    - *Supported by clear criteria and communication plans*
  - In many regulated or safety-critical environments, failing to stop unsafe operations is more serious than temporary downtime

Triggers

- Mark when fallbacks and contingencies activate
  - Fallbacks and contingencies must not depend on arbitrary judgment alone
  - Triggering conditions should be predefined, risk-based, and operationally clear
  - Many incidents start with attempts at standard mitigation
  - Fallbacks/contingencies kick in when
    - *Mitigation is taking longer than RTO permits*
    - *There is no clear ETA for resolution*
    - *Complex root-cause analysis is ongoing and cannot be rushed*
  - Example
    - *Mitigation for an authentication bug is estimated at 6 hours; RTO is 1 hour*
    - *Trigger fallback (reduced functionality or alternate auth), while mitigation continues in parallel*

Triggers

- KRIs indicate escalating instability
  - KRIs can act as early-warning triggers for fallback/contingency
    - *Rising error rates in a core service beyond dynamic thresholds*
    - *Increasing “near misses” in transaction processing*
    - *Accelerating capacity saturation and queuing delays*
  - Example
    - *KRI: Payment failure rate > 5% and rising*
    - *Trigger fallback: temporarily route payment traffic to alternate payment provider or degrade non-critical functions*

Triggers

- RTO/RPO thresholds are at risk
  - When the time since incident onset approaches RTO
  - Potential data loss is approaching RPO
  - Fallback and contingency must activate to avoid breach of resilience commitments
  - Example
    - *RTO = 2 hours for core banking services*
    - *After 90 minutes of failed mitigation, risk team triggers contingency DR site activation or reduced functionality mode*
  - Here, timeliness is non-negotiable

Triggers

- Customer-facing SLAs are compromised
  - SLAs (latency, uptime, transaction success) often define
    - *Contractual obligations*
    - *Financial penalties*
    - *Reputational risk thresholds*
  - Triggered when live metrics show SLA breaches or imminent breaches
  - Example
    - *Latency for core API > 2s for more than 10 minutes*
    - *Uptime < 99.9% for the current period*
  - Fallback might
    - *Disable secondary features*
    - *Increase rate-limiting on non-critical clients*
    - *Trigger traffic rerouting to reduce load*

Safety Nets

- High-resilience organizations treat fallback and contingency planning as dynamic
  - Plans are updated after major incidents and Post-Incident Reviews
  - New attack vectors or failure modes result in new fallback/contingency patterns
  - Automation increasingly orchestrates fallback activation (e.g., automatic failover, circuit breakers, traffic shifting)
  - Exercises include forced activation of fallback/contingency to validate feasibility and human readiness

Business Continuity and Disaster Recovery

- Business continuity and disaster recovery (BC/DR)
  - Ensures critical services and business functions can withstand, adapt to, and recover from disruptive events
    - *Including technical failures, cyberattacks, third-party outages, physical incidents, or large-scale crise*
  - Business Continuity (BC): Keeps business processes running, even if in degraded mode.
  - Disaster Recovery (DR): Restores technology systems to a trusted, operational state.
    - *BC focuses on what the business must be able to do*
    - *DR focuses on what the technology must be able to provide to support BR*

Adaptive Systems

- Self-monitoring: continuous telemetry and anomaly detection
  - Systems continuously emit and analyze telemetry such as
    - *Latency, error rates, queue depth*
    - *Authentication patterns*
    - *Resource usage, saturation metrics*
    - *Configuration and policy drift*
  - Self-monitoring means
    - *Metrics and logs are first-class citizens*
    - *Anomaly detection (statistical and ML-based) is built-in*
    - *Signals are correlated across services and layers*

Adaptive Systems

- Self-healing: automated remediation routines
  - Self-healing means automated correction of certain classes of faults
    - *Restarting crashed processes or containers.*
    - *Redeploying from a golden image when configuration drift is detected.*
    - *Replacing unhealthy nodes automatically.*
    - *Revoking tokens or disabling accounts detected as compromised.*
  - Self-healing reduces MTTR without waiting for human intervention

Adaptive Systems

- Self-adjusting: auto-scaling, rebalancing, failover
  - Self-adjusting systems
    - *Scale horizontally under load (add more instances/pods)*
    - *Rebalance traffic to healthy nodes or regions*
    - *Shed non-critical load when under stress (graceful degradation)*
    - *Adjust concurrency limits and rate limits dynamically*
  - Examples
    - *Auto-scaling groups in cloud environments that increase capacity in response to CPU or response-time signals*
    - *Service meshes that reroute traffic around failing instances*
    - *These behaviors reduce the likelihood that a surge or partial failure escalates into a full outage*

Adaptive Systems

- Machine-assisted reasoning
  - AI/ML-assisted threat classification
  - Adaptive systems leverage AI/ML to
    - *Classify alerts based on historical data and context*
    - *Score risk levels for events or entities (e.g., user risk scores)*
    - *Identify subtle patterns of attack (low-and-slow or multi-stage threats)*
    - *Suggest likely root causes and recommended actions*
  - These capabilities augment human teams, particularly during high-volume or complex events

Organizational Adaptation

- Cross-functional collaboration
  - Adaptive organizations break silos between
    - *Security*
    - *Operations / SRE*
    - *Development*
    - *Risk and GRC*
    - *Legal and Compliance*
    - *Business owners*
  - Response involves multi-disciplinary teams working from a shared situational picture, not separated by rigid boundaries

Organizational Adaptation

- Rapid, evidence-based decision-making
  - Decisions are
    - *Data-informed (based on KPIs, KRIs, telemetry, threat intel)*
    - *Time-bounded (decisions made in minutes, not days)*
    - *Guided by pre-agreed frameworks (risk appetite, playbook criteria)*
  - Adaptive organizations invest in
    - *Clear decision rights*
    - *Predefined thresholds*
    - *Real-time dashboards*

Organizational Adaptation

- Flattened communication channels
  - During high-severity events
    - *Hierarchical communication slows response*
    - *Adaptive organizations empower direct, lateral communication across teams*
  - This may involve
    - *Dedicated incident channels*
    - *Real-time incident “war rooms”*
    - *Clear but lightweight approval chains*

Organizational Adaptation

- Dynamic allocation of staff and expertise
  - Adaptive response requires
    - *Quickly pulling in experts from across the org*
    - *Reprioritizing non-critical work to free key people*
    - *Using on-call rotations and swarm models effectively*
  - Organizations may maintain
    - *Incident commander rosters*
    - *Domain expert lists (e.g., IAM, payments, data)*
    - *Surge capacity plans for major incidents*

Organizational Adaptation

- Continuous learning from incidents
  - Every incident (and near miss) becomes input into
    - *Updated playbooks*
    - *Better KRIs and KPIs*
    - *Architecture improvements*
    - *Training and simulations*
  - Move from repeating the same failures to building resilience through learning

Examples of Adaptive Response

- Auto-isolation of compromised endpoints
  - Endpoint detection and response (EDR) tool detects suspicious behavior
    - *For example: ransomware-like encryption*
  - Automatically isolates the endpoint from the network
  - Triggers SOAR workflows: ticket creation, alerting, forensic snapshot
- Risk impact
  - Limits lateral movement
  - Containment is immediate, reducing blast radius

Examples of Adaptive Response

- Automated throttling of suspicious network activity
  - Anomaly detection identifies spikes in traffic from certain IPs or geos, or unusual query patterns
  - System automatically throttles or blocks traffic according to predefined policies
  - Security and Ops teams are alerted to validate and tune behavior
- Risk impact
  - Reduces DDoS or data-exfiltration impact
  - Protects critical services while investigation proceeds

Examples of Adaptive Response

- Continuous reprioritization of risk remediation
  - Based on real-time KRIs
  - KRI dashboards highlight rising exposures (e.g., vulnerability clusters, failed controls, near misses)
  - Risk teams continuously adjust remediation queues and priorities
  - Automated workflows reassign tasks, adjust due dates, and notify owners
- Risk impact
  - Ensures mitigation efforts target the most critical and time-sensitive risks, not just what is “on the plan”

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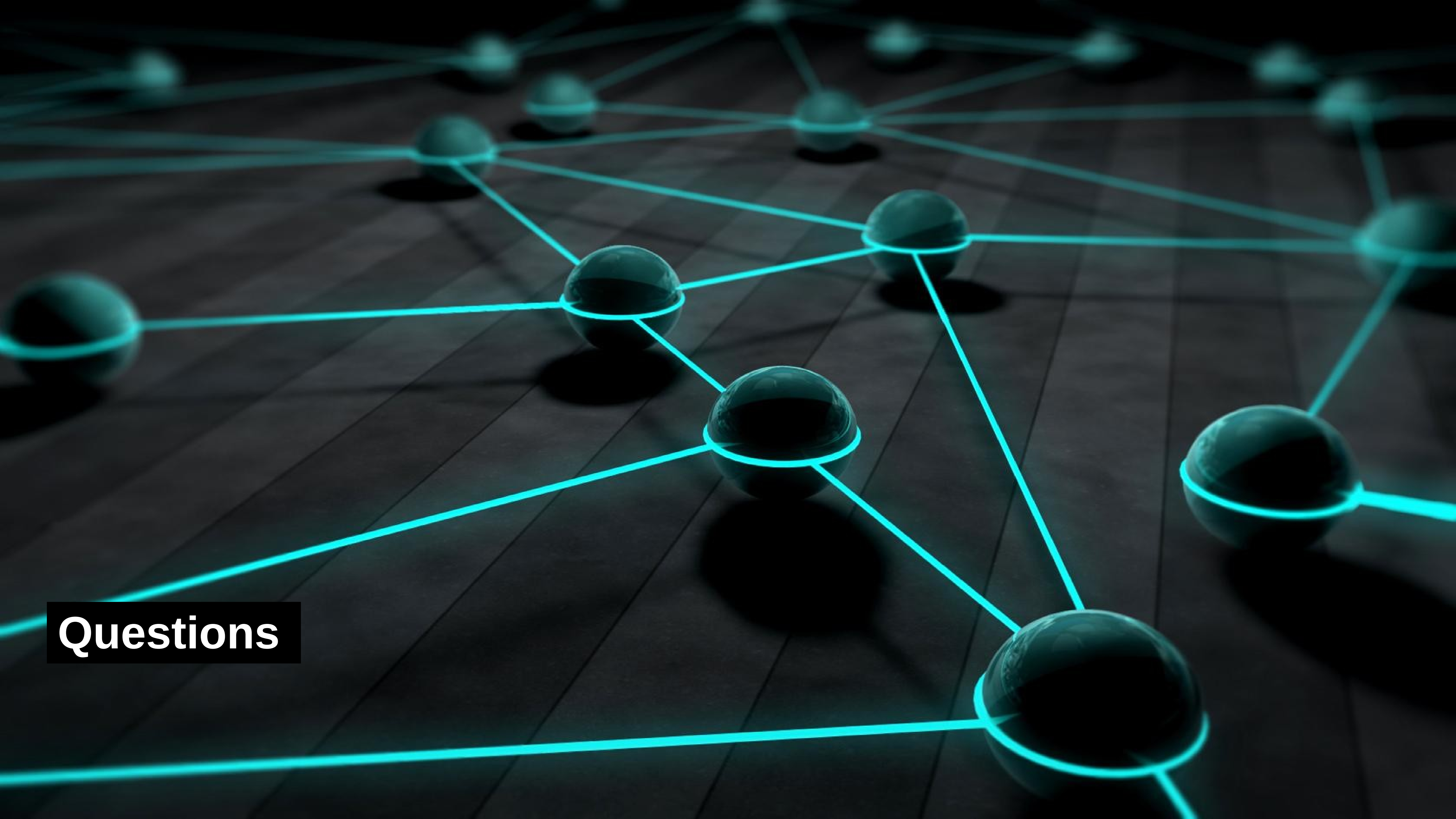

Questions